

Project Proposal-Group04

Project Name:

TalentVerify AI – Intelligent Candidate Screening and Professional Profile Verification System

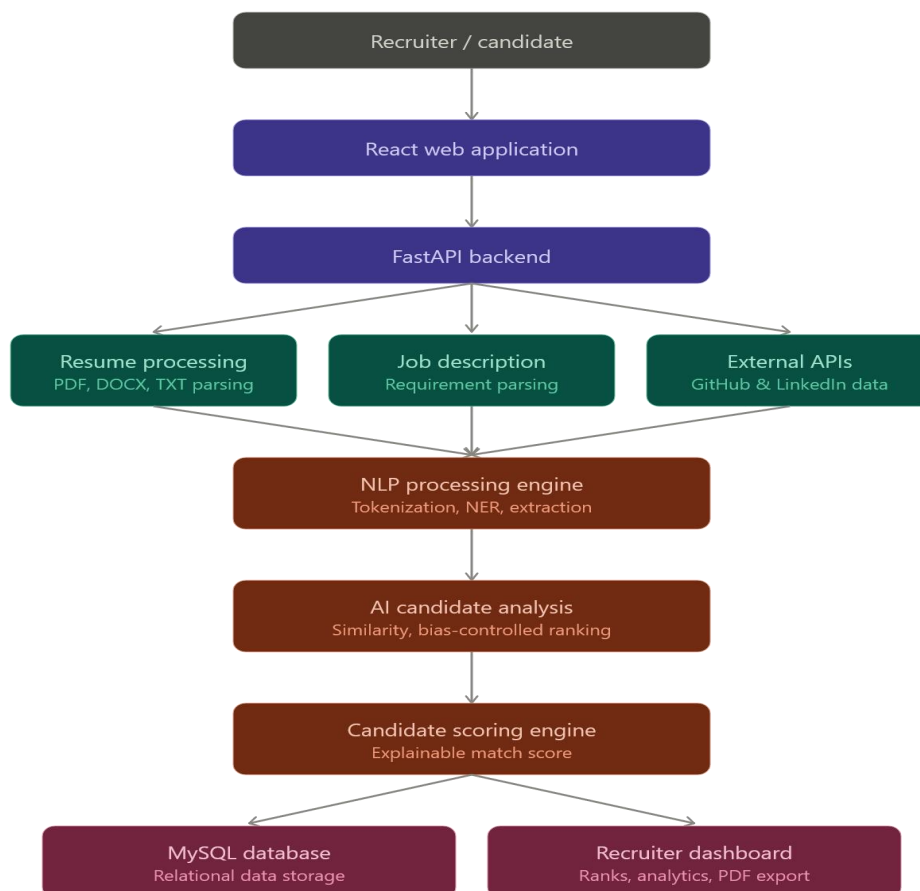
1. Overview

TalentVerify AI is an AI-powered candidate screening and ranking system that uses Natural Language Processing (NLP) and Machine Learning to analyse resumes and match candidates with job requirements. With the candidate's consent, the system also evaluates GitHub and LinkedIn profiles to verify skills and professional experience. It generates explainable scores, ranks candidates, identifies skill gaps, and provides recruiters with accurate and efficient hiring recommendations.

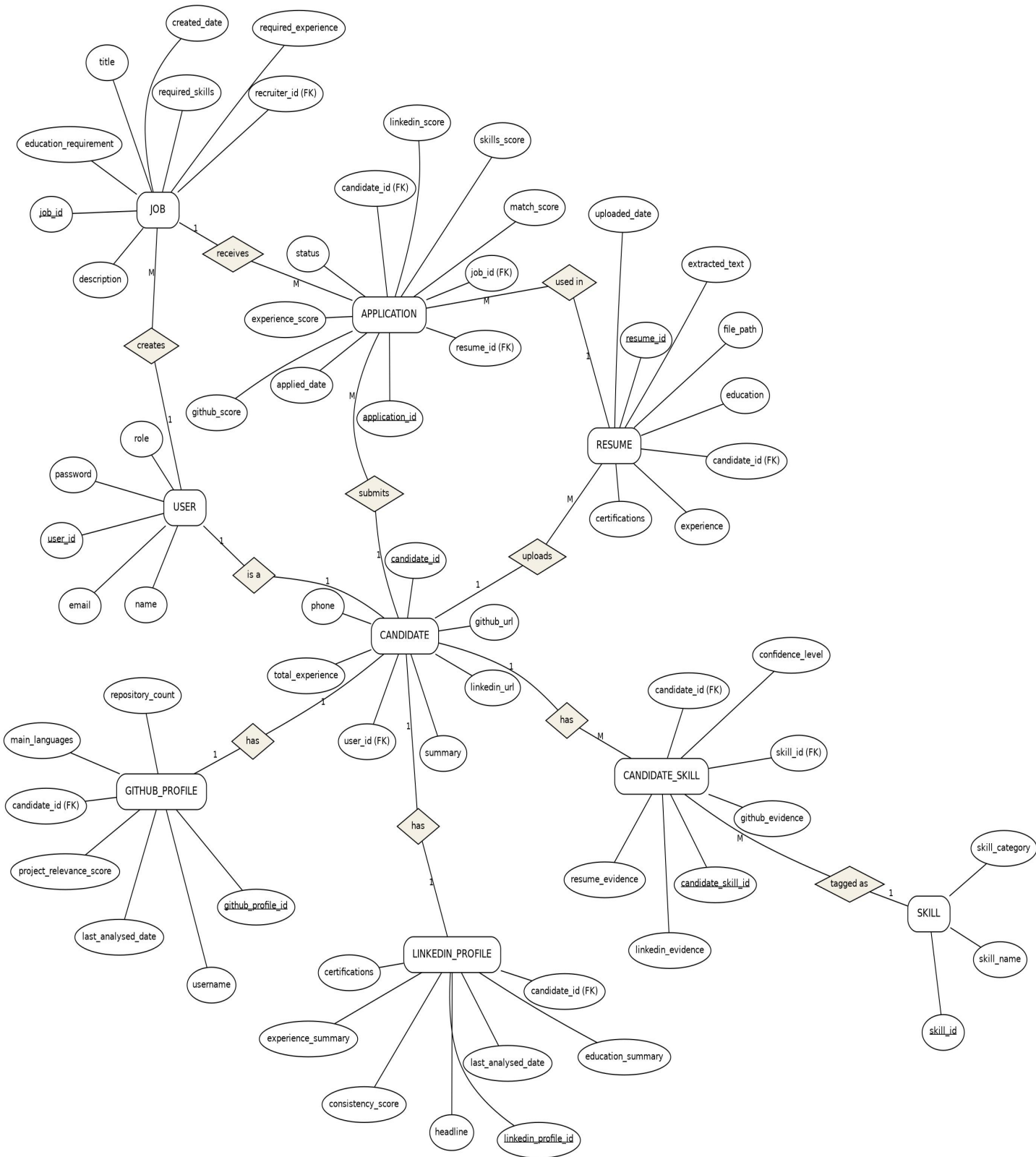
2. Challenges Intended to Solve Using AI Integration

- Slow Resume Screening Process
- Inaccurate Skill Claims
- Difficulty in Candidate Comparison
- Bias in Recruitment Decisions
- Lack of Transparent AI Results

3. System Architecture Diagram



4.System ER Diagram



5. Functional Requirements

- Candidate Functions, Recruiter Functions, AI Functions

Non-Functional Requirements

s-Performance, Security, Privacy, Fairness and Explainability, Maintainability

6. Technology Details

Frontend -React, HTML5, CSS3, JavaScript

Backend -Python, FastAPI, REST API, JWT Authentication

AI and NLP -spaCy, Sentence-BERT, Scikit-learn, TF-IDF, Transformer Models, SHAP Explainability

Resume Processing -PyMuPDF, python-docx, Regular Expressions

Database -MySQL

External Integration-GitHub REST API, LinkedIn Authorised API, OAuth 2.

Deployment -Docker, GitHub Actions, AWS / Azure / Render

7. Conclusion

TalentVerify AI provides an intelligent and explainable recruitment solution by combining Artificial Intelligence, Natural Language Processing, Machine Learning, and multi-source candidate verification.

The system reduces recruitment time, improves candidate comparison, identifies unsupported skill claims, and provides transparent AI-based recommendations. Privacy, fairness, and human oversight will be maintained throughout the system.

This project is suitable for an AI-based Web Technologies course because it integrates AI, web development, databases, API integration, security, and cloud deployment.

9. GitHub Repository Link

<https://github.com/Erangaisuru22/talentverify-ai>